

Research on the Intrinsic Drivers, Realistic Dilemmas, and Development Pathways of Digital Intelligence-Empowered Transformation of Skill-Oriented Higher Education Institutions

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Abstract: With the rapid advancement of AI, big data, and IoT, industrial transformation has entered a new phase of digital intelligence, raising higher competency demands for talent. As the primary base for cultivating high-skilled workers, skill-oriented higher education institutions face both opportunities and challenges for systemic change. This paper analyzes the drivers, dilemmas, and pathways of digital intelligence-empowered transformation from three perspectives. The study identifies three intrinsic drivers: industrial upgrading as demand-side pull, national policies and contemporary development as environmental traction, and institutional system change as organizational adaptation. Realistic dilemmas include internal adaptive obstacles among institutions, teachers, and students, as well as external structural imbalances in university-enterprise and inter-institutional relationships. To address these, a four-dimensional collaborative pathway—concept, competence, relationship, and institution—is proposed, integrating digital intelligence into educational values, bridging competency gaps, empowering relational networks, and implementing institutional safeguards. This study offers systematic theoretical and practical guidance for the digital intelligence transformation of skill-oriented higher education institutions.

Keywords: *Digital Intelligence; Skill-Oriented Higher Education Institutions; Intrinsic Drivers; Realistic Dilemmas; Development Pathways*

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1. Introduction

With the rapid iteration of digital intelligence technologies (AI, big data, IoT), the digital economy has become a global engine. As industrial transformation moves toward digital intelligence, higher competency demands for talent have emerged. China's digital economy faces a talent gap of over 32 million, especially severe in high-end compound skilled talents [1]. This creates an urgent need for transformation of skill-oriented higher education institutions.

At the policy level, the Chinese government has issued multiple documents. In 2024, nine departments released the Action Plan for Accelerating the Cultivation of Digital Talents (2024-2026), aiming to build a high-level digital talent team [2]. The 2025 Opinions on Data Element Disciplines and Digital Talent Cultivation called for a mechanism driven by technological development and national strategies [3]. In 2026, the Ministry of Education issued Opinions on Deepening Key Elements Reform in Vocational Education Teaching, covering majors, curricula, textbooks, teachers, and internship training [4]. These policies provide direction and safeguards for transformation.

Research indicates that high-skilled talent cultivation in skill-oriented higher education institutions is shifting toward high-end innovation, digital competence, open collaboration, and dynamic evaluation [5]. However, in the context of deep digital intelligence empowerment, the transformation has gone beyond simple technology implantation or equipment upgrading, evolving into a systemic change involving educational philosophy, governance, cultivation systems, and evaluation mechanisms. Therefore, this paper systematically explains the intrinsic drivers, realistic dilemmas, and development pathways of digital intelligence-empowered transformation, aiming to provide theoretical reference for relevant reform practices.

2. Intrinsic Drivers of Digital Intelligence-Empowered Transformation of Skill-Oriented Higher Education Institutions

2.1 Demand-Driven: Paradigm Shift in Industrial Upgrading and Skill Talent Requirements

The accelerating pace of technological iteration is a major driving force prompting the transformation of skill-oriented higher education institutions. Technologies such as artificial intelligence, the Internet of Things, and digital twins are evolving rapidly, and industrial requirements for skilled talents are continuously upgrading [5]. Currently, China is in a critical period of transitioning from a manufacturing giant to a smart manufacturing powerhouse. The acceleration of digital, networked, and intelligent upgrading of traditional industries, coupled with the continuous emergence of new industries, has profoundly impacted the demand structure for skilled talents.

First, there is the coexistence of a technical skill gap and employment pressure. On one hand, enterprises face recruitment difficulties, particularly in finding suitable high-skilled talents. On the other hand, university graduates face employment difficulties, especially lacking the practical operation capabilities required by the market. This contradiction highlights a significant skill supply-demand imbalance between talent cultivation in skill-oriented higher education institutions and actual industrial needs. Therefore, the digital intelligence transformation of skill-oriented higher education institutions is driven by both industrial demand and quality improvement imperatives, serving as the engine powering the institutions' own development and their response to social needs.

Second, there is the timeliness challenge of significantly shortened skill renewal cycles. Rapid technological iteration has greatly shortened the shelf life of skills. According to IBM research, the skill half-life in the information technology field has shortened to 2-3 years, and in some cutting-edge technology fields, it is even less than 2 years [6]. This means that the professional knowledge and skills students acquire during their studies may already be partially outdated before graduation. This places demands on skill-oriented higher education institutions for continuous updating and agile response. For example, professional offerings must adapt dynamically to the development of industrial clusters, teacher competencies must be continuously

iteratively updated, and course content must keep pace with technological developments, rather than using the same textbook for ten years. If institutions fail to adjust their training programs in a timely manner, students may face the risk of unemployment upon graduation.

Third, there is the compound shift of fundamentally changing skill demand structures. Traditional skilled talents primarily met the needs of standardized, repetitive positions. However, the digital intelligence era requires compound skilled talents who can base themselves on professional skills and digital literacy, extend their capabilities through equipment operation and data analysis, and ultimately achieve integration of production processes and collaborative management systems. Research shows that against the backdrop of technological advancement in the intelligent manufacturing industry, the competency structure of high-skilled talents is rapidly upgrading from single-operation type to diverse compound type. The demand gap for compound high-skilled talents is most prominent among enterprises, and cultivating such talents is also the most difficult [5].

2.2 Environmental Traction: The Coupling Effect of National Policies and Contemporary Development

The national level has elevated the cultivation of digital talents to a strategic height, and the sustained efforts of policy systems constitute an important traction force for the transformation of skill-oriented higher education institutions [2]. In terms of policy tool configuration, current digital talent policies have formed a combination of mandatory, incentive, and capacity-building tools. In terms of policy content implementation, the Opinions on Deepening the Reform of Key Elements in Vocational Education Teaching by the Ministry of Education takes five key elements—majors, curricula, textbooks, teachers, and internship training—as entry points to systematically reshape the operational logic of industry-education integration. Among these, the professional dynamic adjustment mechanism emphasizes using big data to accurately predict talent supply and demand and implementing a red-yellow card prompt system, providing a clear and operable policy pathway for skill-oriented higher education institutions to solve the problem of disconnection between professional offerings and industrial needs [4].

From the perspective of contemporary development background, the comprehensive implementation of the Digital China strategy provides a grand historical opportunity for the transformation of skill-oriented higher education institutions. As the pace of

industrial digital transformation accelerates, enterprise requirements for skilled talents have risen from basic tool operation to comprehensive competencies in data application, collaborative management, and innovative practice [1]. Facing this change, skill-oriented higher education institutions must base themselves on the talent cultivation requirements of the digital intelligence era, construct a new governance and education paradigm centered on data, relying on multi-dimensional data collection and intelligent analysis to promote precision in talent supply-demand matching and resource allocation. They should build human-machine collaborative virtual-real integrated teaching models and establish full-cycle personalized and flexible evaluation and certification systems to adapt to the practical needs of high-skilled talent cultivation under digital intelligence transformation [5]. Only in this way can skill-oriented higher education institutions find their proper positioning in serving digital economy development and demonstrate their irreplaceable value.

2.3 Organizational Adaptation: Endogenous Requirements for Systemic Change in Skill-Oriented Higher Education Institutions

The survival and development pressures, quality improvement imperatives, and organizational change needs of skill-oriented higher education institutions themselves constitute endogenous driving forces for digital intelligence transformation.

First, there is the practical need to resolve survival and development crises. With the decline in the working-age population and the increasing popularization of higher education, competition for student enrollment among higher education institutions is becoming increasingly fierce. Those skill-oriented higher education institutions that are unable to establish deep connections with industry and have low graduate employment quality may face systemic disadvantages in terms of student enrollment, employment, and social reputation. At the same time, non-traditional educational providers such as corporate universities and online education platforms are entering the vocational education field, impacting skill-oriented higher education institutions. At a deeper level, this competition is not only about student enrollment but also about institutional legitimacy. When the training efficiency and quality of corporate-run education surpass those of traditional institutions, the value foundation for the existence of skill-oriented higher education institutions will be shaken. Therefore, it is urgent for skill-oriented higher education institutions to accelerate their transformation.

Second, there is the inevitable choice to respond to quality improvement imperatives.

According to research, local higher education institutions generally face difficulties such as weak capacity to serve enterprises and lack of distinctive educational characteristics [7], and the same applies to skill-oriented higher education institutions. The root cause lies in the systemic deviation between talent cultivation quality and industrial needs. Digital intelligence transformation offers an opportunity to break this deadlock. Through data-driven precision teaching, personalized learning support, and industry-education integrated practical training, students' job competence and employment competitiveness can be effectively enhanced. Admittedly, digital intelligence technology itself does not necessarily bring quality improvement. Technology is merely a tool; the real driving force for change comes from whether the institution is willing to break established interest patterns and operational habits and translate student-centeredness from a slogan into actionable teaching behaviors.

Third, there is the requirement for leading systemic organizational restructuring. Digital intelligence transformation is not partial repair or technological overlay, but a comprehensive systemic restructuring involving educational philosophy, governance structure, training models, evaluation mechanisms, and more [8]. The current transformation difficulties of many skill-oriented higher education institutions do not stem from insufficiently advanced technology, but from the inability of institutions to adapt. For example, a simple curriculum adjustment may require multiple layers of approval and take months, while the skill changes demanded by industry may occur in just a few weeks. In the era of digital intelligence technology empowerment, skill-oriented higher education institutions must break away from previous fragmented reform thinking and adopt a systemic perspective to promote holistic change from philosophy to institutions, from technology to culture.

3. Realistic Dilemmas of Digital Intelligence-Empowered Transformation of Skill-Oriented Higher Education Institutions

3.1 Internal Ecology: Adaptive Obstacles of Multiple Actors

First, the traditional institutional operation models of skill-oriented higher education institutions have not yet effectively adapted to the requirements of digital intelligence. From the institutional perspective, the realistic dilemmas faced by skill-oriented higher education institutions in their transformation are mainly concentrated in four

aspects: rigid school-running models, outdated talent training programs, disconnection between curriculum systems and job requirements, and low intensity of platform construction aggregation. In terms of school-running models, most institutions still center on disciplinary knowledge systems, pursuing comprehensiveness in professional offerings but showing significantly insufficient sensitivity and response speed to changes in industrial demand. This operational habit of disciplinary logic overriding application logic leads to a disconnect between institutional supply and industrial demand. For example, in training programs, fixed academic systems and uniform pacing still predominate, lacking modular and flexible design, with clear gaps between teaching content and the digital skills required by jobs. Regarding curriculum systems and job requirements, students at skill-oriented higher education institutions generally believe that course content is organized according to disciplinary knowledge logic rather than job competency mapping as the logical starting point, and the speed at which new technologies, processes, and standards enter the curriculum is far slower than the pace of industrial technology iteration, with teaching content lagging behind actual enterprise needs. Regarding platform construction, with the rapid development of digital intelligence technologies such as artificial intelligence, most institutions lack localized deployment capabilities for new technologies as well as unified planning and intensive construction mechanisms at the institutional level. At the root, the deep-seated crux of these dilemmas lies in the lack of systematic top-level design for digital intelligence transformation at the institutional level.

Second, teachers' digital literacy lags behind the requirements of digital intelligence teaching transformation. In the context of artificial intelligence increasingly empowering education, many teachers have not yet formed systematic and complete ideas for applying digital intelligence tools. Teachers are key implementers of digital intelligence transformation, but there is a significant competency gap between their digital literacy and transformation requirements. On one hand, teachers have insufficient ability to integrate emerging digital intelligence tools such as generative AI, intelligent teaching assistants, and adaptive learning platforms. Most still use traditional teaching models, lacking the design thinking to deeply embed digital intelligence technologies into the entire process of intelligent pre-class preparation, dynamic in-class interaction, and precise post-class evaluation. On the other hand, teachers' data literacy does not match the requirements of data-driven precision teaching. The core of digital intelligence teaching lies in the collection and analysis of

multimodal learning behavior data to achieve personalized guidance. However, most teachers are not adept at using digital intelligence tools for learning situation mining, learning path recommendation, or teaching decision optimization. Research shows that the digital literacy of teachers in higher vocational colleges faces realistic dilemmas such as simple assimilation of digital technology and vocational skills, lack of digital teaching research and innovation capabilities, and singleness of knowledge and skill reserves [9]. Furthermore, the teacher digital literacy cultivation system has not yet responded in a timely manner to the needs of the digital intelligence era. Existing training mostly remains at the level of tool operation, lacking systematic cultivation of deeper competencies such as digital intelligence-empowered instructional design and human-machine collaborative teaching strategies. There is a structural mismatch between training content and teachers' real needs. What teachers lack is not an operation manual, but a thinking framework and innovative pathways for integrating digital intelligence technologies into instructional design [10].

Third, the subjectivity of students in digital intelligence learning has not been effectively stimulated. Artificial intelligence is rapidly penetrating learning scenarios, yet most students have not yet developed the autonomous learning abilities and critical thinking habits adapted to this context. Students are the ultimate beneficiaries of digital intelligence transformation, but there is a certain gap between their learning methods and the core requirements of the digital intelligence era for talent, mainly manifested as strong passive learning habits, weak independent inquiry abilities, and shallow use of digital intelligence tools. On one hand, students are accustomed to passively receiving information, lacking the awareness to actively explore and deeply process. When faced with answers provided by generative AI, many students are satisfied with using them as-is, without verifying information sources, questioning the logic of generation, or reflecting on the boundaries of conclusions. For example, when completing course assignments or project reports, students often directly copy AI-generated content rather than using it as a starting point for thinking or reference material, resulting in the learning process degenerating into simple information transfer. On the other hand, students' use of digital intelligence tools remains at the level of shallow collection rather than deep mining and knowledge construction. They can use digital intelligence tools to quickly retrieve information and organize literature abstracts, but rarely ask about the reasons behind the data, the relationships between viewpoints, or the assumptions underlying conclusions. Research indicates

that students face a crisis of subjectivity in the digital intelligence era. In intelligent educational environments, students easily degenerate from active learners into passive recipients, and their critical thinking and creative abilities face the risk of being weakened [8]. At the same time, students' subject status has not been fully manifested. The essence of digital intelligence technology empowering education is to unleash human potential, not replace human thinking. If students lose their learning initiative and judgment in the digital intelligence environment, such technology application deviates from the true purpose of education.

3.2 External Ecology: Structural Imbalances in Multi-dimensional Relationships

In the development process of digital intelligence-empowered transformation of skill-oriented higher education institutions, the multi-dimensional relationships in the external ecology exhibit relatively significant structural imbalances, mainly manifested in the following two aspects.

First, industry-education integration remains superficial, lacking a bidirectional empowerment mechanism. From the institutional perspective, schools expect to enrich their talent cultivation forms through the input of advanced enterprise technologies and the implementation of real projects. However, enterprise-level participation is mostly in shallow forms such as achievement demonstrations and equipment donations. From the enterprise perspective, enterprises in industry-education integration practice also urgently need skill-oriented talents cultivated by schools, but the degree of fit between the curriculum content students learn in higher education institutions and the technologies required by industrial job positions is not high. In summary, cooperation can only be equal when schools can prove their technological service value. Without sustainable long-term benefit-sharing mechanisms, university-enterprise cooperation may remain superficial.

Second, inter-institutional competition intensifies, making resource sharing difficult to implement. At the individual institutional level, under the multiple factors of decreasing total student enrollment, limited funding allocation, and narrowing policy dividends, higher education institutions are competing for favorable positions in various institutional rankings. Generally, institutions are unwilling to openly share their core advantageous resources. This situation leads to increasingly fierce competition for student enrollment and prominent phenomena such as redundant curriculum development. From the perspective of the overall operational effectiveness

of higher education institutions, this fragmented situation reduces the efficiency of educational resource utilization. Some scholars point out that although various university alliances have been established successively, many alliances in practice are trapped at the level of temporary initiatives, with some scholars noting that although many alliances had many visions at their establishment, they ultimately degenerate into information exchange platforms, and the internal motivation for sustained operation gradually diminishes [11]. Therefore, when every institution pursues becoming the ultimate winner as its primary goal, the resource allocation efficiency of the entire system tends to decline. Only when each institution proactively breaks down resource barriers and adopts an open attitude to achieve complementary advantages and mutual learning can sustainable impetus be injected into the digital intelligence transformation of skill-oriented higher education institutions, more effectively responding to the urgent demands for high-quality skilled talents raised by contemporary development.

4. Development Pathways for Digital

Intelligence-Empowered Transformation of Skill-Oriented Higher Education Institutions

4.1 Integrating Digital Intelligence Technologies into Educational Values

Against the backdrop of digital intelligence empowerment becoming the core driving force for industrial upgrading, the transformation and upgrading of skill-oriented higher education institutions, as the main front for cultivating high-skilled talents, has become an inevitable move to adapt to contemporary development and meet industrial demands. Exploring scientific and feasible development pathways is key to ensuring the effective implementation of digital intelligence transformation in skill-oriented higher education institutions. The digital intelligence transformation of skill-oriented higher education institutions must first address the value premise of why to transform. The current dilemmas of rigid institutional operation models and disciplinary logic overriding application logic at the institutional level stem from the failure to update educational philosophy in step with contemporary transformation. In terms of institutional positioning, skill-oriented higher education institutions need to shift from discipline-based to competency-based, from pursuing comprehensive disciplinary

layouts to creating differentiated advantages based on regional industrial needs. Research shows that the cultivation of high-skilled talents in the digital intelligence era is evolving from skill adaptation to skill boundary-crossing and skill iteration [5]. This means that skill-oriented institutions can no longer use the standardized thinking of the industrial age to address the personalized needs of the digital intelligence era; they should establish professional adjustment mechanisms that dynamically adapt to the development of industrial clusters. In terms of cultivation philosophy, digital technology is not the key to educational change. True digitization involves applying digital technology to achieve educational model innovation based on changes at the cognitive and conceptual levels of thinking [12]. Therefore, digital intelligence empowering vocational education is not about making classrooms more dazzling, but about enabling students to acquire the key abilities and core competencies needed to adapt to the digital intelligence era. This requires institutions to take job competency mapping as the logical starting point for curriculum design, reverse-engineer the curriculum system, and ensure precise alignment between teaching content and industrial needs. It must be understood that digital intelligence technology is merely a means, while talent cultivation is the purpose. Whether skill-oriented higher education institutions can achieve high-quality transformation and development through digital intelligence empowerment ultimately depends on whether the institutions can adhere to the value concept of serving both student development and industrial needs.

4.2 Using Digital Intelligence Technologies to Bridge the Missing Competencies of Main Actors

Teachers' lagging digital literacy and students' lack of learning subjectivity are core bottlenecks restricting the deepening of transformation. To address these dilemmas, a dual-actor competency reconstruction system covering both teachers and students needs to be constructed. At the teacher level, there needs to be a shift from technical training to systemic empowerment. Addressing the problem of teachers' insufficient ability to integrate emerging tools such as generative AI and intelligent teaching assistants, institutions should construct a tiered and categorized teacher digital competence development system, incorporating digital intelligence teaching capability into teacher evaluation and professional title promotion systems. However, current training mostly remains at the level of tool operation, lacking systematic cultivation of deeper competencies such as digital intelligence-empowered instructional design and human-machine collaborative teaching strategies. Therefore,

institutions should promote the transformation of teacher training from surface-level tool application to deep-level enhancement of core competencies, strengthening systematic learning of modules such as digital intelligence instructional design and intelligent teacher-student collaboration strategies. Training should not stop at providing user manuals, but should help teachers form a thinking framework and innovative pathways for integrating digital intelligence technologies into the teaching process.

At the student level, there needs to be a shift from passive acceptance to active construction. Addressing the problem of students being accustomed to passively receiving information and lacking critical thinking about AI-generated content, institutions should promote student-centered blended teaching reform, establishing process-based evaluation and precise intervention mechanisms based on learning behavior data. Research shows that using academic early warning models for early identification and precise assistance of academically struggling students can reduce failure rates by nearly 20% [12]. At the same time, students should be guided to shift from shallow use of digital intelligence tools to deep knowledge construction, for example, learning to question information sources, verify generation logic, and reflect on the boundaries of conclusions. The essence of digital intelligence technology empowering education is to unleash human potential, not replace human thinking. If students lose their learning initiative and judgment in the digital intelligence environment, such technology application deviates from the true purpose of education.

4.3 Using Digital Intelligence Technologies to Empower New Types of Relational Networks

At the level of school-enterprise relations, schools need to shift the cooperation from one-way taking to two-way empowerment, and establish a sustainable mechanism for shared benefits. On one hand, schools should continuously provide high-quality skilled talents to enterprises; on the other hand, they should proactively offer technical services, employee training, technological research and development, etc., as reverse empowerment. For example, schools can build physical cooperation platforms such as industrial colleges and industry-academia integration bases, and explore deep cooperation models such as shareholding systems and mixed ownership. At the inter-school level, schools should shift the relationship from zero-sum competition to symbiosis and prosperity. On one hand, they need to actively break down resource

barriers and build substantive cooperation carriers such as cross-school course sharing platforms, virtual teaching rooms, and credit mutual recognition mechanisms, and reduce the cost of repeated construction through the joint establishment of digital teaching resource libraries. On the other hand, based on their own advantages, they should achieve differentiated development and inject continuous impetus into the digital transformation of technical colleges through complementary advantages. At the inter-institutional relationship level, institutions should promote the shift of inter-institutional relationships from zero-sum game to symbiosis and shared prosperity. On one hand, they need to proactively break down resource barriers, building substantive cooperation carriers such as cross-institutional curriculum sharing platforms, virtual teaching and research sections, and credit mutual recognition mechanisms, reducing redundant construction costs through the co-construction of digital teaching resource libraries. On the other hand, they should achieve differentiated development based on their own advantages, injecting sustainable impetus into the digital intelligence transformation of skill-oriented higher education institutions through complementary advantages.

4.4 Using Digital Intelligence Technologies to Effectively Implement Institutional Safeguards

The sustainability of digital intelligence-empowered transformation of skill-oriented higher education institutions ultimately depends on the adaptability of the institutional system. Traditional bureaucratic management systems are overly rigid and insufficiently flexible, conflicting with the agile response required by digital intelligence transformation, which needs to be resolved through systematic institutional innovation.

At the evaluation mechanism level, institutions should move from single standards to diversified evaluation. Some scholars point out that the evaluation system for Double First-Class construction has drawbacks such as emphasizing papers and rankings with single standards, leading many institutions to lose direction in the competition for first-class status [7], which in this case can easily deviate from the requirements of technical skill talent cultivation. In response, empowered by digital intelligence technologies, skill-oriented higher education institutions can establish new data-driven, multi-actor participating, continuously improving evaluation systems. For example, in teacher evaluation, shift from publication-oriented to multi-contribution orientation, incorporating teaching reform, curriculum development, technical

services, and transformation into the evaluation system. In student evaluation, shift from outcome evaluation to a combination of process and outcome evaluation, incorporating process data such as class participation, project performance, and teamwork ability into academic evaluation. In program evaluation, introduce industry enterprises and third-party institutions to participate, taking employer satisfaction as a core indicator of talent cultivation quality.

At the data governance level, institutions should move from fragmented silos to comprehensive integration. Addressing the problem of institutions lacking unified planning and intensive construction mechanisms at the institutional level, a three-tier data governance architecture of collection—governance—service can be established, implementing the principle of one data source. Through data governance, three major goals can be achieved: precise online data, cross-departmental collaboration, and data value realization [13]. Institutions should establish an authority responsible for coordinating university-wide information technology construction, formulating mandatory data standards and interface specifications. In fact, the insufficient intensity of platform construction aggregation appears on the surface to be a technical management problem, but is actually a shortfall in the institution's digital governance capability. If the fragmented construction pattern cannot be broken at the institutional level, digital intelligence transformation will remain stuck in a low-level, fragmented repetition stage for a long time.

At the policy tool level, a shift from command guidance to incentive empowerment is needed. Current policy tools, though diversified, lack sufficient incentive mechanisms, leading to a risk of policy suspension—institutions feel pressure but lack motivation [14]. National and local levels should increase the proportion of incentive tools (e.g., special funds, project support, tax incentives) to stimulate endogenous motivation of institutions, teachers, and enterprises. The core issue is not policy quantity but the low fit between policy design and institutional needs; without reaching vital interests, policies remain on paper. Internally, institutions should establish fault-tolerant mechanisms to avoid the reverse incentive where more work leads to more blame.

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